HUNT-SAGE · THREAT HUNTING CHEAT SHEET

LOLBins Quick Reference

Signed Windows binaries attackers abuse to download, execute, dump and persist, and the telemetry that gives each one away.

github.com/iamjammie/hunt-sage · iamjammie.github.io/hunt-sage · Built 2026-11-03

Living catalogues: **LOLBAS** (Windows), **GTFOBins** (Linux), **LOOBins** (macOS), **LOLDrivers** (vulnerable/malicious drivers), **LOLRMM** (abused remote-management tools), **WADComs** (AD/Windows offensive tooling, for mapping to artefacts).

Attackers prefer signed, Microsoft-shipped binaries because they are trusted, pre-installed and often ignored by baseline allow-lists. The binary name alone is almost never the signal. Hunt on the combination:

binary + arguments + parent process + integrity level + network + user + path

A `rundll32.exe` launched by `services.exe` loading a System32 DLL is normal; the same binary launched by `winword.exe`, loading a DLL from `%TEMP%` and making an outbound connection, is not.

Download and ingress transfer

Binary	Legitimate use	Detection signals	ATT&CK
`certutil.exe`	Certificate/CA management	`-urlcache`, `-split`, `-verifyctl`, `-f` with an `http(s)` URL; `-encode` / `-decode` turning files to/from base64	T1105, T1140
`bitsadmin.exe`	Background file transfer	`/transfer`, `/addfile`, `/create`, `/resume`; `/SetNotifyCmdLine` chaining to another binary (also persistence)	T1197, T1105
`curl.exe`	HTTP client (built in since Win10 1803)	`-o` / `-O` writing to `%TEMP%`, `%APPDATA%`, `ProgramData`; upload with `-T` / `--upload-file`; non-browser user agent to a rare host	T1105, T1048
`powershell.exe` / `pwsh.exe`	Automation	`Invoke-WebRequest`, `iwr`, `Invoke-RestMethod`, `.DownloadString(`, `.DownloadFile(`, `Start-BitsTransfer`; often with `-nop -w hidden -enc`	T1105, T1059.001
`mshta.exe`	Run HTML applications	Remote `http(s)://` or `javascript:` / `vbscript:` argument (fetches and runs in one step)	T1218.005, T1105
`esentutl.exe`	ESE database utility	`/y` copy from a URL or UNC path; `/vss` to read locked files	T1105, T1003.003
`desktopimgdownldr.exe`	Lock-screen image fetch	Registry value set under `PersonalizationCSP` pointing at a remote URL, then the binary run	T1105
`finger.exe`	Finger protocol client	Output redirected to a file; used as a tiny TCP downloader to port 79	T1105
`expand.exe` / `extrac32.exe`	Expand CAB archives	Source path is a UNC share or URL; writing to a user-writable path	T1105
`replace.exe` / `print.exe`	Replace/print files	Source is a UNC or remote path; `print /D:` writing an arbitrary file	T1105, T1570

Execution proxies

Binary	Legitimate use	Detection signals	ATT&CK
`rundll32.exe`	Run DLL exports	DLL from `%TEMP%` / `%APPDATA%`; `javascript:`; `comsvcs.dll, MiniDump`; no arguments then network; odd `ordinal` exports	T1218.011, T1003.001
`regsvr32.exe`	Register COM DLLs	"Squiblydoo": `/s /n /u /i:` with an `http(s)` URL and `scrobj.dll`	T1218.010
`mshta.exe`	Run HTA	Parent is Office, a browser or mail; remote content argument	T1218.005
`wscript.exe` / `cscript.exe`	Windows Script Host	`.js`, `.vbs`, `.wsf`, `.vbe` launched from `Downloads` / `%TEMP%` / `%APPDATA%`; Office or archive tool as parent	T1059.005, T1059.007
`msiexec.exe`	Install MSI packages	`/q /i` with an `http(s)` URL or a UNC path; MSI from a user-writable path	T1218.007
`installutil.exe`	.NET installer	`/logfile= /LogToConsole=false /U` with a DLL from a user path (runs the uninstall method)	T1218.004

Binary	Legitimate use	Detection signals	ATT&CK
`regasm.exe` / `regsvcs.exe`	.NET COM registration	`/U` with a DLL from a user-writable path	T1218.009
`msbuild.exe`	Build .NET projects	A `.csproj` / `.xml` / `.proj` under a user path; MSBuild making network connections or spawning `cmd` / `powershell`	T1127.001
`cmstp.exe`	Connection Manager profiles	`/s` / `/ns` with an `.inf` from a user path (also an auto-elevated UAC bypass)	T1218.003, T1548.002
`control.exe`	Control Panel	A `.cpl` (really a DLL) loaded from a user-writable path	T1218.002
`odbcconf.exe`	ODBC configuration	`/a` with `{REGSVR <dll>}` ; `-f` response file registering a DLL	T1218.008
`wmic.exe`	WMI command line	`process call create` ; `/node:` remote exec; `/format:` with a remote XSL (`.xsl`)	T1047, T1220
`forfiles.exe` / `pcalua.exe`	Batch over files / compat assistant	`/c cmd /c` or `-a` / `-p` spawning another binary (indirect execution hides the real parent)	T1202
`conhost.exe`	Console host	Used as a launcher prefix to break the parent/child chain	T1202
`scriptrunner.exe`	App-V script runner	`-appvscript` launching an arbitrary command	T1218
`mavinject.exe`	App-V DLL injection	`/INJECTRUNNING <pid>` with a DLL path (injects into a live process)	T1055.001
`xwizard.exe`	Extensible wizards	`RunWizard` with a CLSID, loading a DLL from the working directory	T1218
`msdt.exe` / `pcwrun.exe`	Diagnostics	External/URL protocol invocation (the "Follina" pattern); unexpected parent	T1218
`verclsid.exe`	Verify shell extension	Launched with a CLSID to approve a COM object out of the usual Explorer flow	T1218.012
`ieexec.exe` / `presentationhost.exe`	.NET/ClickOnce hosting	Loading a remote `.application` or `.xbap` /managed app	T1127

Credential and memory access

Binary	Legitimate use	Detection signals	ATT&CK
`rundll32.exe` + `comsvcs.dll`	COM services	`comsvcs.dll, MiniDump <pid> <file> full` against `lsass.exe`	T1003.001
`rdrleakdiag.exe`	Resource-leak diagnosis	`/memdump` / `/fullmemdmp` targeting the LSASS pid	T1003.001
`createdump.exe`	.NET crash dump	`-f <file> -u <pid>` or targeting `lsass.exe`	T1003.001
`sqldumper.exe`	SQL Server dump	`<lsass-pid> 0 0x01100:40` style full-process dump of LSASS	T1003.001
`werfault.exe` / `WerFaultSecure.exe`	Windows Error Reporting	Dumping LSASS via a forced error or `-u -p <pid>`	T1003.001
`tttracer.exe`	Time-Travel Debugging	`-dumpFull -attach <lsass-pid>`	T1003.001
`ntdsutil.exe`	AD database maintenance	`"activate instance ntds" "ifm" "create full"` run off a DC or by non-admins	T1003.003
`reg.exe`	Registry editor	`save` / `export` of `HKLM\SAM` , `HKLM\SYSTEM` , `HKLM\SECURITY`	T1003.002
`vssadmin.exe` / `diskshadow.exe`	Shadow copies	`create shadow` then reading `ntds.dit` / hives from the snapshot	T1003.003

Discovery and reconnaissance

Binary	Detection signals	ATT&CK
`net.exe` / `net1.exe`	`group "domain admins" /domain` , `user /domain` , `localgroup administrators` , `session` , `view` , `use \\host\ipc$`	T1087, T1069, T1135
`nltest.exe`	`/dclist:` , `/domain_trusts` , `/all_trusts` , `/dsgetdc:`	T1482, T1018
`dsquery.exe` / `dsget.exe`	`* -filter` , `-f "(objectcategory=...)"` , large result sets	T1087.002
`whoami.exe`	`/all` , `/priv` , `/groups` (quick token/privilege triage)	T1033

Binary	Detection signals	ATT&CK
`systeminfo.exe`, `hostname.exe`, `ver`	Host and patch-level enumeration early in a session	T1082
`tasklist.exe` / `qprocess.exe`	`/v`, `/svc`, `/fi` filtering for security products	T1057, T1518.001
`quser.exe` / `qwinsta.exe` / `query.exe`	Who else is logged on, RDP session enumeration	T1033
`ipconfig`, `arp`, `route`, `netstat`, `nbtstat`	Network and neighbour discovery burst from one host	T1016, T1049
`reg.exe query` / `wevtutil el`	Reading config, listing event logs before tampering	T1012, T1082
`fsutil.exe`	`fsutil fsinfo`, `usn` queries; enumerating volumes	T1082

A single one of these is noise. A **burst of many distinct discovery binaries from one host within minutes**, especially from a non-admin or a server that does not normally run them, is the hunt. Stack `(parent, child)` pairs and `dcount` of discovery binaries per host per hour.

Persistence and remote execution

Binary	Detection signals	ATT&CK
`schtasks.exe`	`/create` with `/ru SYSTEM`, an action in a user path, or `/s \\host` for remote; hidden tasks	T1053.005
`sc.exe`	`create ... binPath=` to a user path or `cmd /c`; `config ... start= auto`; remote `\\host`	T1543.003
`reg.exe`	`add` to `...\Run`, `RunOnce`, `Winlogon\Shell` / `Userinit`, `Image File Execution Options`, service `ImagePath`	T1547.001, T1546.012
`dnscmd.exe`	`/config /serverlevelplugindll <path>` (loads a DLL into the DNS service on a DC)	T1543.003
`at.exe`	Legacy scheduled jobs (deprecated but still present on older hosts)	T1053.002
`wmic.exe` / PowerShell CIM	`__EventFilter` / `CommandLineEventConsumer` WMI subscription creation (Sysmon 19/20/21)	T1546.003
`pwsh` / `powershell` profile writes	New `profile.ps1`, `Microsoft.PowerShell_profile.ps1` in user or all-users paths	T1546.013

Staging, archiving and copy

Binary	Detection signals	ATT&CK
`makecab.exe` / `cabarc.exe`	Compressing collected files before exfil, often in `%TEMP%`	T1560, T1074
`tar.exe` (built in)	`-c` creating an archive of a data directory; `.tar` / `.gz` in a staging path	T1560.001
`xcopy.exe` / `robocopy.exe` / `copy`	Bulk copy to `\\host\C$` / `ADMIN$` or a staging folder; `/mir`, `/s` across shares	T1570, T1074
`findstr.exe`	`/s /i password` recursively over config files; reading from a UNC share	T1552.001, T1105
`certreq.exe`	`-post` uploading a file to a remote URL (small exfil channel)	T1048

Impact and recovery inhibition

Binary	Detection signals	ATT&CK
`vssadmin.exe`	`delete shadows /all /quiet`, `resize shadowstorage`	T1490
`wbadmin.exe`	`delete catalog`, `delete systemstatebackup`, `delete backup`	T1490
`bcdedit.exe`	`/set {default} recoveryenabled no`, `bootstatuspolicy ignoreallfailures`	T1490
`wmic.exe`	`shadowcopy delete`	T1490
`cipher.exe`	`/w:` wiping free space (anti-forensic / destruction)	T1485, T1070.004
`fsutil.exe`	`usn deletejournal`, `file setzerodata`	T1070, T1485

Defense evasion and log tampering

Binary	Detection signals	ATT&CK
`wevtutil.exe`	`cl <log>` / `clear-log` (clearing Security, System, PowerShell logs)	T1070.001
`PowerShell`	`Clear-EventLog`, `Limit-EventLog`, `Remove-EventLog`; `Set-MpPreference` disabling Defender	T1070.001, T1562.001
`sc.exe` / `net stop`	Stopping EDR, Defender, WinDefend, Sysmon or the event-log service	T1562.001
`takeown.exe` / `icacls.exe`	Taking ownership / granting rights on security tooling or system files	T1222.001
`attrib.exe`	`+h +s` hiding dropped files	T1564.001
`fltmc.exe`	`unload` of an EDR/AV minifilter driver	T1562.001
`MpCmdRun.exe`	`-RemoveDefinitions -All`, `-DownloadFile` (Defender's own binary abused)	T1562.001, T1105

UAC bypass and elevation (auto-elevated binaries)

Auto-elevating binaries spawn a High-integrity child after a hijacked registry handler is written. The signal is a **registry write to `HKCU\Software\Classes\...\shell\open\command` or `...\ms-settings\...`** immediately followed by the binary running and a Medium-to-High integrity transition.

Binary	Hijack path to watch	ATT&CK
`fodhelper.exe`	`HKCU\Software\Classes\ms-settings\Shell\Open\command`	T1548.002
`computerdefaults.exe`	`ms-settings` handler	T1548.002
`sdclt.exe`	`App Paths\control.exe` / `Folder\shell\open\command`	T1548.002
`eventvwr.exe`	`mscfile\shell\open\command`	T1548.002
`wsreset.exe`	`AppX...\Shell\open\command`	T1548.002
`slui.exe`	`exefile\shell\open\command`	T1548.002

See the UAC bypass playbook.

Tunnelling and proxy

Binary	Detection signals	ATT&CK
`ssh.exe` (built in)	`-R` / `-L` / `-D` creating reverse, local or dynamic tunnels; outbound to a non-corporate host	T1572, T1090
`netsh.exe`	`interface portproxy add` (port forwarding); `advfirewall` rule changes to allow C2	T1090, T1562.004
`OpenSSH sshd`	A listening `sshd` appearing on a workstation that should not run one	T1572

Detection helpers

- **`OriginalFileName`** (Sysmon 1, Defender `ProcessVersionInfoOriginalFileName`): survives renaming, so a `svchost.exe` on disk whose `OriginalFileName` is `rundll32.exe` is caught.
- **Path**: these ship in `System32` / `SysWOW64`. A copy running from `Users`, `%TEMP%`, `ProgramData`, `Public` or a mounted ISO is rarely legitimate.
- **Parent/child**: Office apps, browsers, mail clients, `wmiprvse.exe`, `w3wp.exe`, `wsmprovhost.exe`, `mshta`, `wscript` as parents of interpreters or the binaries above are high-signal (see the LOLBin proxy execution playbook).
- **Network from the unusual**: `rundll32`, `regsvr32`, `mshta`, `msbuild`, `certutil`, `installutil` making outbound connections (Sysmon 3, `DeviceNetworkEvents`).
- **Integrity transition**: a Medium-integrity parent producing a High-integrity child via an auto-elevated binary.
- **Command-line obfuscation**: carets (`c^e^r^t^u^t^i^l`), mixed case, `%COMSPEC%`/env-var substring tricks, excessive length, and switch variants (`-` vs `/` vs the en dash `–`). In Sigma the `|windash` modifier covers the switch variants; for base64 use `|base64offset|contains`.

Hunting one-liners

Stack rare binary-from-unusual-path across the fleet (Defender XDR):

```
DeviceProcessEvents
| where Timestamp > ago(7d)
| where FileName in~ ("rundll32.exe","regsvr32.exe","mshta.exe","certutil.exe","installutil.exe","msbuild.exe")
| where FolderPath !startswith @"C:\Windows\System32" and FolderPath !startswith @"C:\Windows\SysWOW64"
| summarize Hosts = dcount(DeviceName), Cmds = make_set(ProcessCommandLine, 5) by FileName, FolderPath
| order by Hosts asc
```

LOLBin making a network connection (Splunk, Sysmon TA):

```
index=sysmon (EventCode=1 OR EventCode=3)
| transaction ProcessGuid maxspan=5m
| search EventCode=3 Image IN ("*\\rundll32.exe","*\\regsvr32.exe","*\\mshta.exe","*\\certutil.exe","*\\msbuild.ex
e")
| stats count values(DestinationHostname) as dest by Computer Image
```

Renamed LOLBin (Sysmon `OriginalFileName` mismatch):

```
index=sysmon EventCode=1
| eval base=lower(mvindex(split(Image,"\\"),-1))
| where OriginalFileName!="" AND lower(OriginalFileName)!=base AND lower(OriginalFileName)!=base.".mui"
| table _time Computer User Image OriginalFileName CommandLine
```

Related: LOLBin proxy execution · PowerShell abuse · Process injection · Query index

HUNT-SAGE
iamjammie/hunt-sage